# Traceability Supplement

TABLE XIV
PROPOSED, UNMEASURED SOURCE-DERIVED TRACEABILITY DIMENSIONS

| ID | Required correspondence | Unit numerator / denominator |
|---|---|---|
| TR01 | identity/activity binding to task, exact artifact revision/digest, activity, and actor | complete slots / required slots |
| TR02 | request/input → rule version → evaluated check/result → decision | complete chains / inventoried decisions |
| TR03 | decision → observed state or artifact effect with before/after evidence | complete correspondences / decisions requiring outcome |
| TR04 | triggering failure → corrective package → exact revalidation → result | complete repair chains / actual repair attempts |
| TR05 | forward requirement → artifact → verification and reverse in-scope change mapping | report forward and reverse separately |
| TR06 | non-author independent exercise of declared audit/replay target | exercised targets / declared targets |

The instrument is proposed and unmeasured; no score was collected or validated. Data and Code Availability identifies the codebooks and source rationale.

## I. PROPOSED TRACEABILITY INSTRUMENT (NOT MEASURED)

The proposed instrument assesses requirement/artifact/action/outcome correspondence, not correctness, hidden reasoning, developer skill, or player benefit. For each dimension $d$, freeze an arm-neutral inventory $U_d$ and the required links $L_d(u)$ for each unit $u$. Let $N_d$ count applicable units in $U_d$, and $n_d$ count units with every required link verified. Independent assessors award 1 only when exact source/version, raw-evidence relations, and locator are verified; otherwise they award 0 and record missing, inaccessible, stale, contradicted, or unverified evidence. Report $n_d/N_d$ only with a complete inventory and $N_d > 0$, alongside disagreement and seeded-corruption controls; do not form weighted composites. Mark incomplete inventories INDETERMINATE and report observed counts and the gap. N/A requires independent evidence of non-applicability. Independently verified zero actual repair attempts can qualify a repair item, but an empty log cannot. Keep PASS, FAIL, NOT_RUN, STAGE_NOT_REACHED, OUT_OF_SCOPE, and MISSING distinct; `stage_not_reached` never passes.